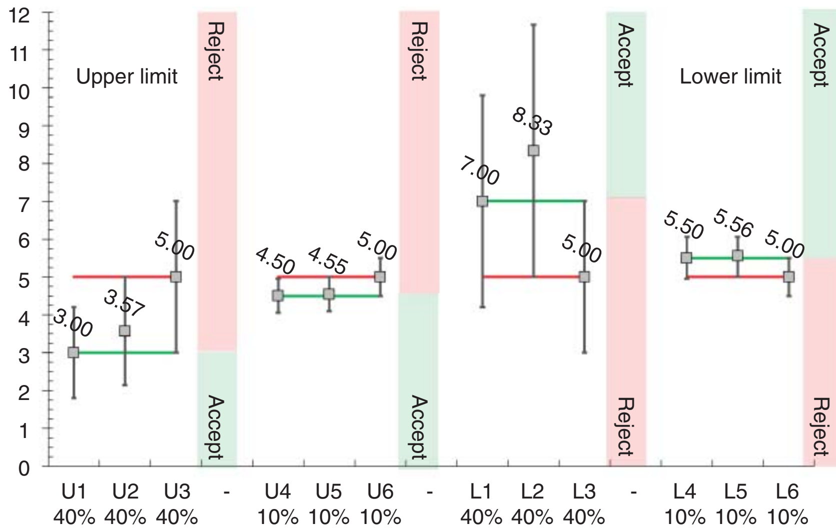

Figure 8.4 Influence of MU on acceptability or rejection intervals. The coverage interval of each measurement value is shown as vertical error bars. Three different situations are represented for each case.

- The size of the guard bands (blue square) depends on the expanded uncertainty.
- The rejection interval is symbolized by a light red rectangle starting from the largest guard band. The acceptability interval is a light green rectangle.

In Figure 8.4, the simulated results of two control methods are illustrated, a well-performing one where the relative uncertainty is $UR\% = 10\%$ and a less efficient one with $UR\% = 40\%$. These percentages are reported below the 12 simulated examples of the figure. Depending on whether the relative uncertainty is 40 or 10%, the guard band is equal to 2.00 or 0.50. It is added or subtracted depending on whether it is a lower or upper specification limit. Corresponding examples are labeled U for the "Upper" limit and L for "Lower."

From there, three situations are considered.

- Cases U1 and U4, L1 and L4. They correspond to the decision rule proposed by the BIPM: the measured content of a sample must be less or equal to the limit of the rejection interval to be in conformity. If the uncertainty is 40% the rejection limit is $5.00 - 2.00 = 3.00$. Thus, case U1 of Figure 8.4 will be declared compliant since its content is precisely 3.00. But when its coverage interval is calculated with the same constant uncertainty of 40% – which is not always the case – it is between [1.80, 4.20]. The upper limit is, therefore, below the specification limit. This sample is penalized.
- This means that a sample presenting a slightly higher value than U1 could give a result less than or equal to 5.00 and therefore be compliant. Furthermore, it is possible to predict what sample can have the upper limit of its coverage interval reaching exactly 5.00.

- Cases U2 and U5 were simulated using this calculation, as described in Section 7.6.4. The reasoning is reversed for cases L2 and L5 in Figure 8.4.
- This effect of relative uncertainty is much less pronounced when it is only 10%.
- Finally, cases U3 and U6 and L3 and L6 show that if the result is equal to the specification limit, 100% of the values included in the coverage interval are in the rejection zone.

According to these diagrams, the four samples numbered 2 and 5 are the most problematic. These samples should be declared compliant since the limit (upper or lower) of their coverage interval corresponds to the specification limit. In other words, only 2.5% of the measurements that could be obtained would not respect the objective. It seems that the principle of the guard band predicted from the expanded uncertainty of the limit of specification applies rather poorly to the analytical sciences for two reasons:

- The uncertainty can be quite high, mainly when close to the LOQ, while regulators try to enforce limits in the vicinity of the LOQ.
- The MU can significantly vary even over a small concentration interval.

It would be more interesting to define the guard band by calculating, from the uncertainty function, at which value the lower or upper bound of the coverage interval is equal to the specification limit, as explained in Section 7.6.4.

This remark underlines the interest of being able to determine beforehand the uncertainty function. In any case, the ISO guide 98-4, which addresses this topic, is mainly oriented toward industrial production, where MU is usually reduced and constant. Indeed, it proposes integrating the statistical characteristics of the production process or other phenomena of interest.

For that, this information must be known *a priori* and described by a probability law, normal, uniform, or other. To account for the variability of the manufacturing process, the guide recommends applying a Bayesian approach which allows integration in the same model of the variability of the production and that of the measurement. But for a service laboratory that receives samples of unknown origin from very varied contexts, the notion of variability due to the manufacturing process is inoperative.

The role of MU in decision-making cannot be discussed without pointing out the reluctance of prescribers to use it. However, with the generalization of the coverage interval, the end-user will be able to benefit from this additional information in a decision-making process or to better interpret a laboratory result. Because the coverage interval contains a probability, it is an indication of the level of confidence a decision maker has to implement a decision rule.

For example, in the context of official food control, the European Commission published a regulation in 2021 that describes how to use MU to enforce *MRL* legislation [10]. As explained in Section 9.3, it results in a new way of calculating two criteria, namely $CC\alpha$ and $CC\beta$, now based on MU that will replace the method proposed in 2002.

8.3 Sampling Uncertainty

8.3.1 Sampling and Heterogeneity

Sampling is a specific field of statistics. There are many textbooks or standards addressing the selection and organization of sampling plans from various technical or industrial perspectives. It is also suggested to refer to standards, regulatory texts, or professional guides or recommendations for defining the best-adapted sampling plan for a given situation. As already pointed out in Chapter 7, a semantic problem should be clarified about the term "sample" widely used in analytical sciences:

- In the laboratory, the sample is the entity or the object on which an analysis is performed. Unfortunately, it may refer to the proper object sent by a user as well as the result of the preparation performed on this object before introduction into the measuring device, such as dilution, solvent extraction, and mineralization.
- In sampling theory, the sample is a set of sampling units taken out of a batch or a population according to some specific rules.

Figure 8.5 illustrates the various terms used in sampling theory to elaborate a sampling plan. Finally, in the customary discussion, the term sample can be used instead of sampling unit, test portion, or the result of its preparation. On the other hand, the term sampling can refer to the operation of collecting a sample or the result of the operation.

To avoid ambiguity, the term sample is used in this chapter to refer to the laboratory meaning of "working sample". According to sampling theory, the number of sampling units depends on the mathematical model chosen to describe the representativeness of the analyte in the situation under study.

In analytical sciences, it is mainly the spatial distribution of the analyte that matters. The general procedure is to produce a statistical model that best describes this spatial distribution and derive the efficiency curve of the sampling plan, i.e. the curve that gives the probability of identifying a nonconforming batch. A sampling plan is

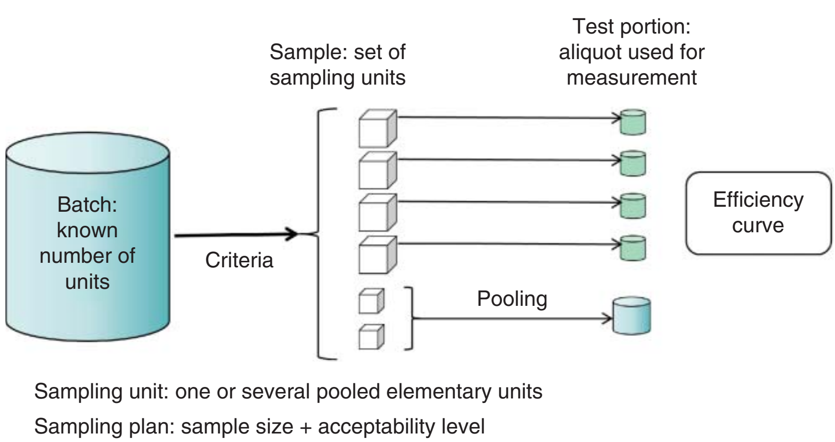

Figure 8.5 Basic sampling vocabulary.

considered efficient when the probability is equal to or above 95%. Many laboratories are not responsible for sampling. Their role is limited to the analytical part of the analysis process, and sampling uncertainty is not under their responsibility. However, for some others, such as clinical laboratories, this is not the case.

When sampling uncertainty is required, the spatial distribution of the analyte is the main source of uncertainty to be accounted for. For instance, sampling uncertainty may be required for the official sanitary control of foods. The spatial distribution relates to homogeneity defined by IUPAC, i.e. "the degree of regularity with which a property or analyte is distributed in a quantity of the material being analyzed." Except for unusual cases, the spatial distribution of the analyte is not regular, either at the lot or sample level and even in liquid products.

Since the 1970s, four types of spatial distribution have been identified and illustrated in Figure 8.6. The so-called uniform distribution is mostly imaginary and does not exist, even after homogenization. This is unfortunate because it is the only case where the sampling uncertainty can be estimated as zero. If the most aggregated distribution is called contagious, it is because it is used as a model in epidemiology to describe the contamination clusters. These graphics show that homogeneity (or heterogeneity) depends on two parameters:

- Spatial distribution of the analyte.
- The ability of the analyte to form aggregates.

Ideally, prior knowledge of these two properties would help optimize the sampling strategy. A theory on sampling uncertainty was postulated by geochemists who introduced the concept of a sampling constant to characterize the spatial distribution of an analyte [11].

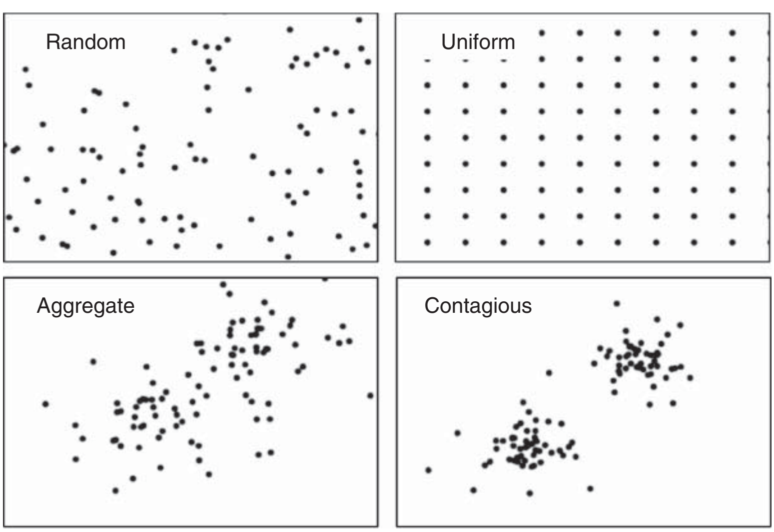

Figure 8.6 Main types of spatial distribution of an analyte in a batch or population.

Many papers were published on this topic because the profitability of the mining industry largely depends on the laboratory's ability to detect nuggets and the beginning or end of a vein. Though the motivation was economic, it paved the way for an essential reflection on sampling.

The proposed model assumes that a lot is formed by a population of N fragments with $1 \leq n \leq N$. The analyte concentration in each fragment is Z_n and contributes to the heterogeneity of the lot by its deviation from the global mean $\overline{Z}$. Each fragment also has its mass m_n and these parameters are related by the following equations:

The total mass of the lot

$$M = \sum_N m_n \tag{8.1}$$

Individual contribution to heterogeneity

$$H_n = \frac{Z_n - \overline{Z}}{\overline{Z}} \times \frac{N \times m_n}{M} \tag{8.2}$$

Global mean

$$\overline{Z} = \frac{\sum_N (m_n \times Z_n)}{M} \tag{8.3}$$

Weighting coefficient

$$W_n = \frac{N \times m_n}{M} \tag{8.4}$$

Weighted variance

$$s_H^2 = \sum_N \frac{(Z_n - \overline{Z})^2 W_n}{\overline{Z}^2 \sum_N W_n} \tag{8.5}$$

These parameters are interpreted as follows:

- When the analyte does not form aggregates, the mean of the H_n is zero, and the variance s_H^2 is used to define a constitutional heterogeneity coefficient HC:

  $$HC = s_H^2$$

- When the analyte forms aggregates, the variance s_H^2 corresponds to a distributional heterogeneity (HD), which is related to the constitutional heterogeneity by the following relation, where FR is a positive aggregation coefficient.

  $$HD = HC \times \frac{1 + FR \times FS}{1 + FR}$$

If $FR = 0$, there is no aggregation, and $HD = HC$.

Parameter FS is a segregation factor, with $1 \geq FS \geq 0$. If $FS = 1$, the segregation is complete, and $HD = HC$. In all other cases, $HD < HC$ and material mixing or homogenizing can reduce its value. From these parameters, a sampling constant can be defined. This is a very elaborate theory mainly inspired by geostatistics, a special field of statistics [12].

The derivation of sampling uncertainty from this model is also complex. It is easily applicable to particulate solids but presents important difficulties for analyzing